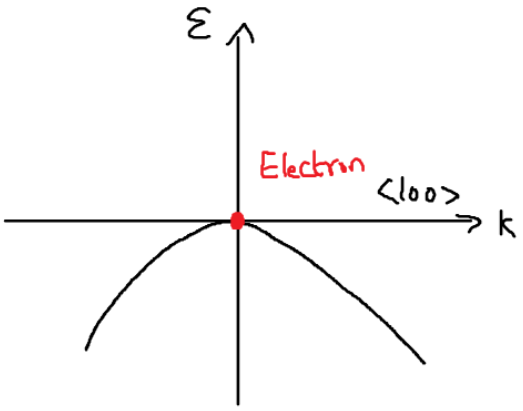
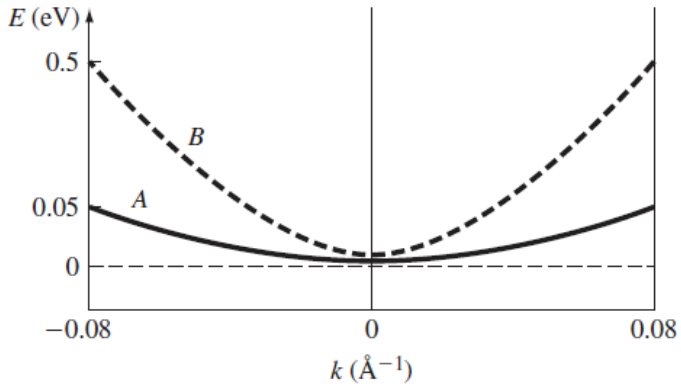
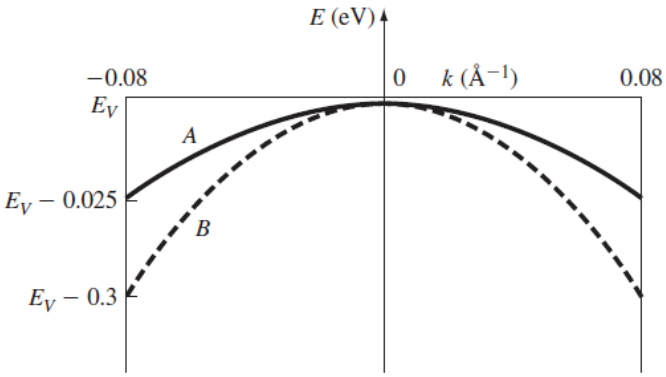
EE2021: Tutorial - 5 (Sept 30th, 2025)

1. Band structure of a hypothetical simple cubic semiconductor at 300 K is shown. It has a lone electron as shown in the symmetrical E-k_x diagram. An electric field is applied in the +x direction. Reason and show initial (dotted circle) and final position (solid circle) in your E-k diagram after a time Δt (assuming t is smaller than mean scattering time)

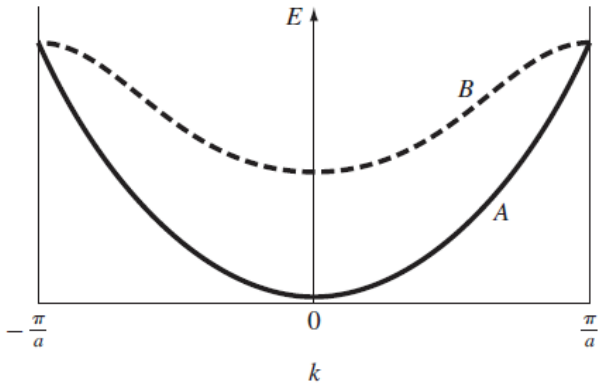


2. Figure below shows the parabolic E versus k relationship in the conduction band (left) and valence band (right) for an electron in two particular semiconductor materials. Determine the relative effective mass (m^*/m_0) for all 4 cases.





3. The E versus k diagrams for a free electron (curve A) and for an electron in a semiconductor (curve B) are shown below. Sketch (a) dE/dk versus k and (b) d^2E/dk^2 versus k for each curve. (c) What conclusion can you make concerning a comparison in effective masses for the two cases?



4. A quantum well LED is made by sandwiching a thin GaAs layer between two $Al_{0.3}Ga_{0.7}As$ layers.The energy band diagram is shown below. Approximate the quantum well for electrons and holes as infinite and find the ground state energy levels in both wells. Find the wavelength of the photon generated by a radiative recombination of electron and hole from these levels. The effective mass of electrons in GaAs is $0.068m_0$ whereas for holes is $0.45m_0$ with m_0 is the free-space mass of electrons.

